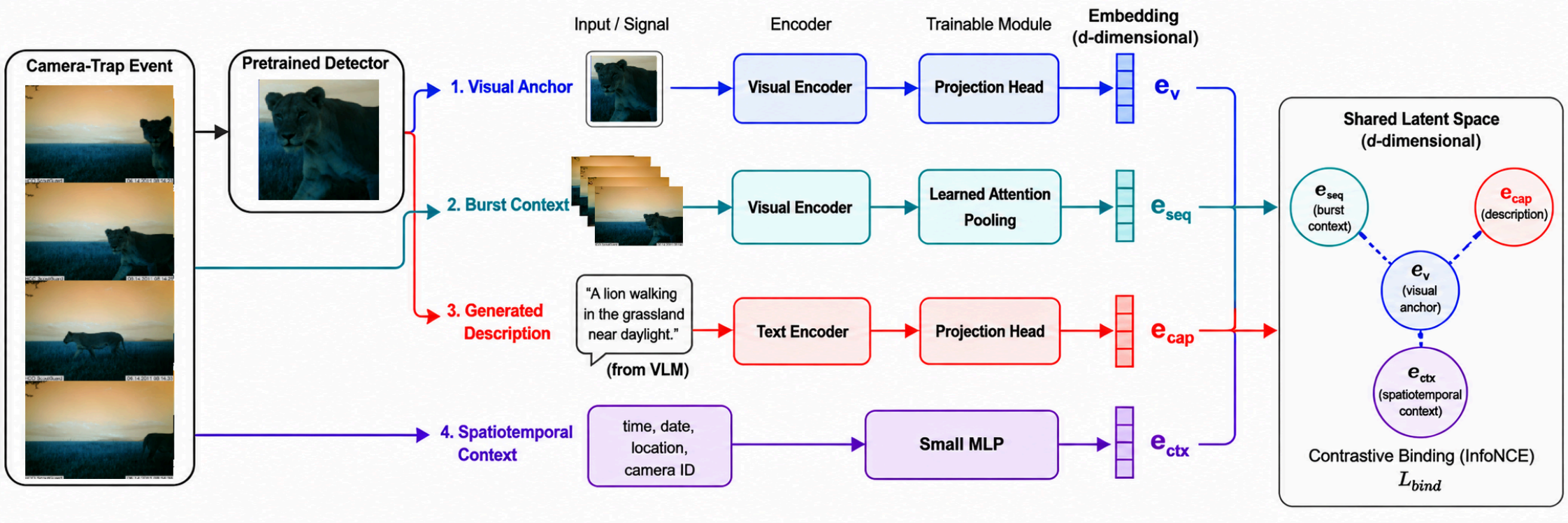

Input / Signal
Encoder
Trainable Module
Embedding
(d-dimensional)
Camera-Trap Event
Pretrained Detector
1. Visual Anchor
Visual Encoder
Projection Head
e_v
2. Burst Context
Visual Encoder
Learned Attention Pooling
e_{seq}
3. Generated Description
"A lion walking in the grassland near daylight."
(from VLM)
Text Encoder
Projection Head
e_{cap}
4. Spatiotemporal Context
time, date, location, camera ID
Small MLP
e_{ctx}
Shared Latent Space
(d-dimensional)
e_{seq} (burst context)
e_{cap} (description)
e_v (visual anchor)
e_{ctx} (spatiotemporal context)
Contrastive Binding (InfoNCE)
L_{bind}